

Mechanic Trust

Closing the Trust Gap in Auto Repair

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Beta prototype · Built with v0 by Vercel · Pre-launch

CONSUMER MOBILE

AI / LLM

AUTO REPAIR

TRUST INFRASTRUCTURE

Type

Consumer mobile app

Stage

Beta, pre-launch

Built with

v0 by Vercel, AI image analysis

Domain

Auto repair / trust

01 · Problem Statement

One-Sentence Problem

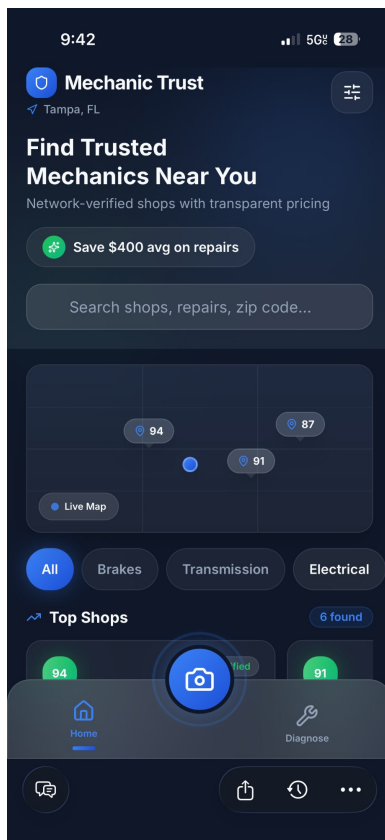
People facing car repairs have no reliable way to find a trustworthy mechanic, understand whether a quoted price is fair, or get a second opinion — leaving them vulnerable to overcharging and bad service with no recourse.

The Deeper Gap

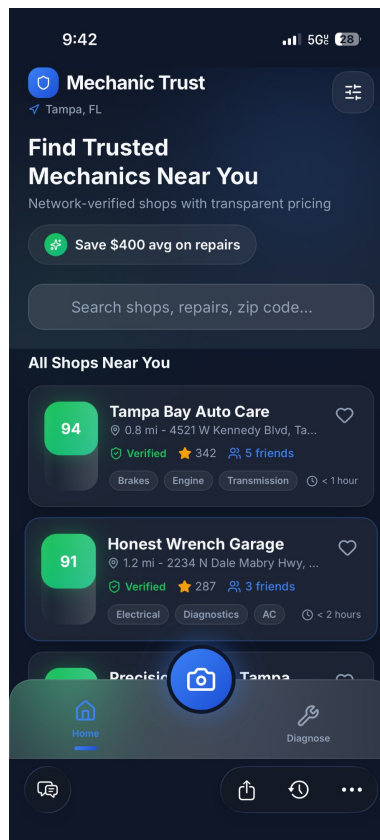
Here's a situation most people have lived through: your car breaks down, you Google "mechanic near me," and you're staring at a wall of 4.5-star reviews that all feel fake. You have no idea if the \$1,400 quote is fair, whether the repair is even necessary, or if the shop is padding the bill because they can tell you don't know better. You're anxious, time-pressured, and making a high-stakes decision with almost zero reliable information.

This isn't niche — **Americans spend over \$80 billion annually** on auto repair. It's one of the largest consumer service categories, and it runs almost entirely on **information asymmetry**: the mechanic knows what's wrong; the driver doesn't. Google and Yelp haven't closed this gap — their ratings are inflated, often gamed, and tell you nothing about whether a shop is competent for *your specific repair* at a *fair price*. And the problem is worsening: modern vehicles — especially hybrids and EVs — are more complex, repair costs are rising, and the pool of qualified independent mechanics is shrinking. The trust gap is getting worse, not better.

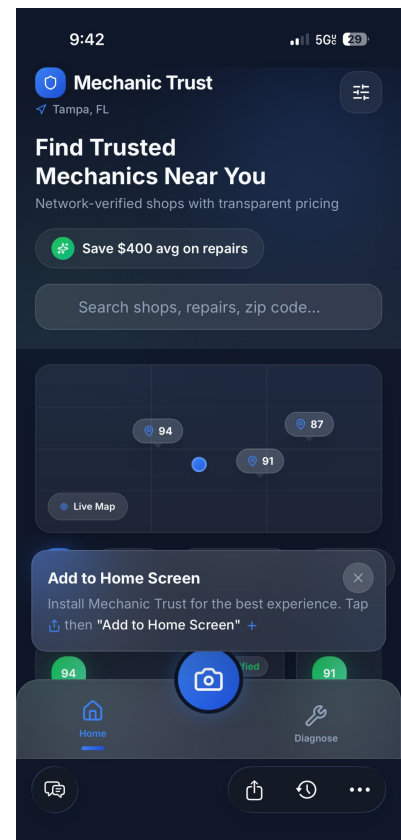
APP SCREENS — DISCOVERY & SHOP FINDING



Home — Find Trusted Mechanics



Top Shops with Trust Scores



PWA Install Prompt

■ ■ 02 · Company Context & Constraints

STAGE

Solo founder, pre-launch beta. No team, investors, or revenue. Built to demonstrate AI PM capability through a production-quality prototype.

TECHNICAL MATURITY

Rapid-prototype stack (v0 for UI). AI image analysis for diagnostics. No live backend or real shop database yet.

RISK TOLERANCE

High velocity, low infra investment. Acceptable to show a working UX with simulated data while deferring live shop API and pricing integrations.

KEY CONSTRAINTS

- Solo build — no eng team
- No live shop-data partnership yet
- AI diagnosis is not mechanically certified
- Social trust layer requires contacts integration (future)

Why This Was the Right Bet

Auto repair is a proven high-anxiety, high-spend category — Americans spend over \$80B annually. Trust-deficit markets are consistently underserved by incumbent review platforms. The social proof layer (friends who've used a shop) is a zero-cost differentiator unavailable on Google or Yelp. Building as a beta let me validate the UX hypothesis — multimodal input + transparent pricing + social trust — before needing live shop partnerships.

■ ■ 03 · Tech Stack & APIs

LAYER	DETAILS
Frontend	v0 → React / Next.js for production. Component-first; fast flow iteration without backend coupling.
AI / LLM	Multimodal LLM (OpenAI or equivalent): confidence score, priority tier (Low/Moderate/High), plain-language assessment, ranked next steps.
Pricing	RepairPal / Mitchell API — price ranges calibrated by repair type + ZIP. Not model inference alone.
Trust & Social	Contacts API (Google / Apple) for 'friends who used this shop.' Trust scores: verification status + review quality + social signal.
Backend	Node / FastAPI service orchestrating AI calls, pricing lookups, contacts matching, trust scoring. Stateless by design.

■ ■ 04 · Key Tradeoffs

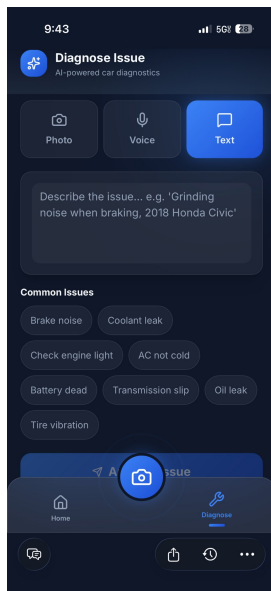
CHOSE TO DO	DID NOT DO	WHY
Rapid UI with v0	Custom React / Next.js	Validate UX flows fast; tech ownership deferred.
AI diagnosis + confidence score	Rule-based diagnostic engine	LLMs handle the long tail better; confidence score keeps output honest.
Price range (\$800–\$1,100) + anchor	Single 'recommended price'	Range is defensible; single number implies false precision.
Social proof via contacts sync	Community review system	No cold-start problem; contacts bootstrap trust from existing relationships.
Multimodal: photo, voice, text	Text-only input	High-stress mobile-first moment; voice + photo cut friction.

■ ■ 05 · Success Metrics

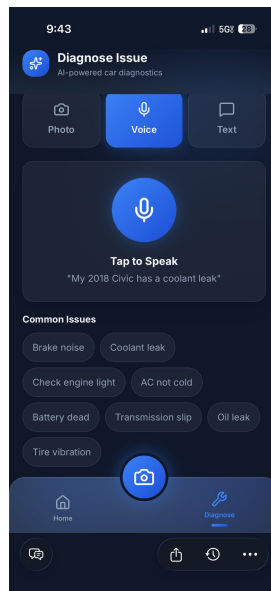
METRIC	TARGET	STATUS	SIGNAL
Diagnosis accuracy	≥75% rated 'useful'	Beta — not yet measured	Defined; test pending
Price range fair	≥70% agree plausible	Needs user testing	Needs user testing
Social proof engagement	20%+ CTR on friend shops	Built; no live data yet	Awaiting contacts API
Full-flow: diagnosis → shop	≥60% in one session	Validated in walkthroughs	Qualitative only
Time to first result	< 60 sec from open	~45 sec manual walkthrough	On track

06 · AI Systems Thinking & Eval Framework

APP SCREENS — DIAGNOSE FLOW



Text Input



Voice Input

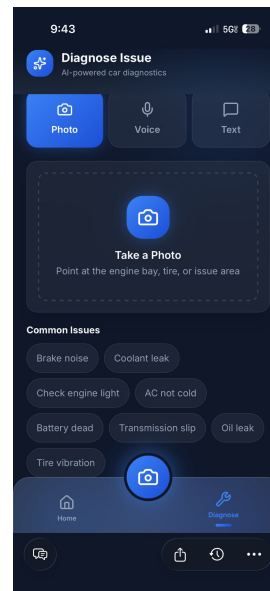
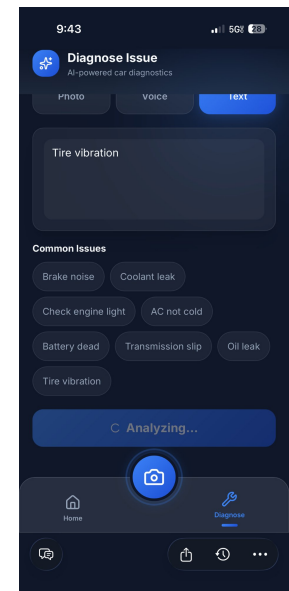
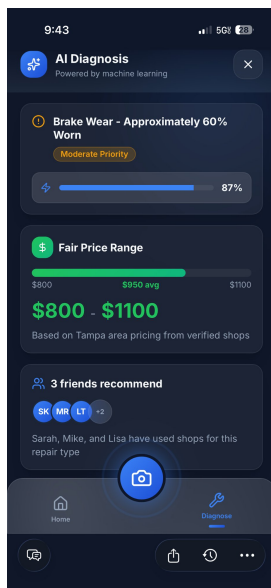


Photo Input

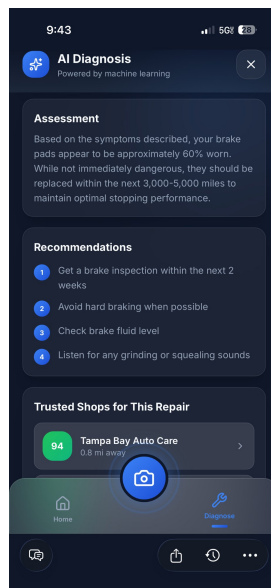


Analyzing...

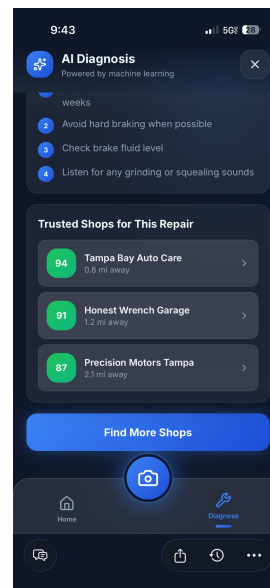
APP SCREENS — AI DIAGNOSIS RESULTS



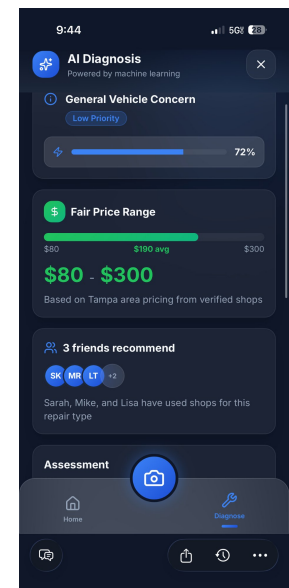
Confidence + Priority



Assessment + Recs



Trusted Shops



Tire Vibration Result

HOW MODEL QUALITY WAS EVALUATED

Quality assessed across three dimensions:

1. **Relevance** — Does the output name a plausible issue?
2. **Calibration** — Does confidence reflect real uncertainty? (Vague photo → lower score.)
3. **Actionability** — Does it produce a useful recommendation and price range, not just a label?

HOW FAILURE MODES WERE DEFINED

Silent failure: Confident-sounding but wrong answer. Mitigation: confidence score always surfaced.

Overconfidence: Serious issue flagged at high confidence on ambiguous input. Mitigation: Moderate vs. High Priority framing with explicit next steps.

Underspecification: Too little user input. Mitigation: common-issue chips pre-populate inputs; voice mode reduces friction.

OBSERVABILITY & GUARDRAIL DECISIONS

Transparency over authority: Never 'your brakes need replacement.' Always 'approximately 60% worn — inspect within 2 weeks.' The AI informs; it does not prescribe.

Source attribution: Price ranges labeled 'based on Tampa-area verified shop pricing,' grounding output in real data.

Priority tiering: Low / Moderate / High with visual distinction prevents alarm fatigue.

No hallucinated shops: Recommendations scoped to verified shops for the specific repair type.

EVAL FRAMEWORK — WHAT I WOULD BUILD NEXT

- Diagnosis acceptance rate — did users book at a recommended shop?
- Price-range accuracy — did actual cost fall within the predicted range?
- Confidence calibration curve — does 80% confidence $\approx$ 80% accuracy?
- Re-diagnosis rate — did users re-submit for the same issue?

■ ■ 07 · Product Judgment Under Uncertainty

JUDGMENT CALL	REASONING
Ship social-proof UI before contacts data	The differentiator requires contacts integration → which requires user trust → which requires a working product. Built UI with simulated signals first. Bet: if UX validates concept, integration is tractable; if UX fails, contacts don't matter.
Show price range, not single estimate	Chose \$800–\$1,100 + \$950 anchor. Every comparable product omits pricing or shows a suspicious single number. A wrong single number destroys credibility; a range the actual quote falls within reinforces trust.
Scope shops to repair type, not geography	All nearby shops = noise. Verified shops for that repair type = smaller list, higher signal. Deliberate IA decision: reduce choice, increase confidence in the options shown.
Three input modes on day one	Someone in a parking lot with a leaking engine is not going to type carefully. Photo, voice, and text cover the full range of use-moment contexts — high-stress, mobile-first.

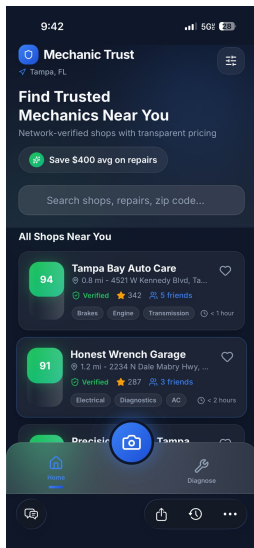
■ ■ 08 · What Would Break If I Was Wrong

ASSUMPTION	WHAT BREAKS IF WRONG
Social proof drives no engagement	Core differentiator collapses. Product becomes a trust-score app — useful but not different from a well-curated Yelp filter.
AI diagnosis perceived as untrustworthy	Users skip Diagnose flow. App becomes a shop-finder — a much thinner product. This is why guardrails are load-bearing, not decorative.
Price range too wide to be useful	\$100–\$2,000 is accurate but useless. Range only anchors expectations if calibrated to local, repair-type, verified-shop data.
Trust score doesn't feel earned	94 vs 87 must feel grounded. If users sense scores are opaque or gameable, the trust infrastructure is undermined — hardest platform risk to rebuild.

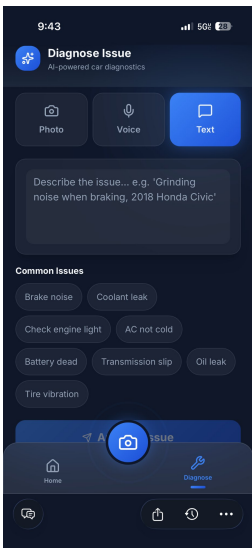
09 · Scope Ownership & Leadership Signal

SCOPE OWNERSHIP 100% solo. Every product decision, UX flow, AI prompt design, trust-scoring logic, and UI architecture — owned and executed by Ruthvik with no team or delegation.	LEADERSHIP IN AMBIGUITY Pre-launch: no users, no pricing-data partnership, no contacts API. Made product bets on the core loop (diagnose → price → book) being sufficient to validate before solving data and integration challenges.
INDEPENDENT PRODUCT JUDGMENT Built against no existing PRD. All prioritization decisions — multimodal input, social proof layer, price-range framing — made independently through user empathy and product reasoning.	AI SYSTEMS THINKING Confidence scoring, failure-mode definition, transparency guardrails, repair-type scoping, and a defined eval framework — applied before a single real user. Pre-market thinking, not post-launch cleanup.

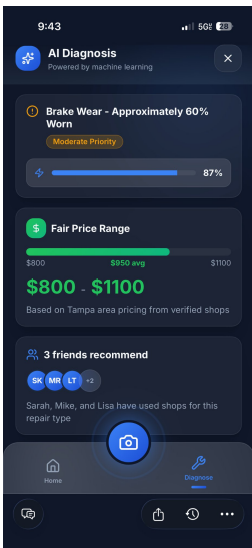
APP SCREENS — COMPLETE FLOW OVERVIEW



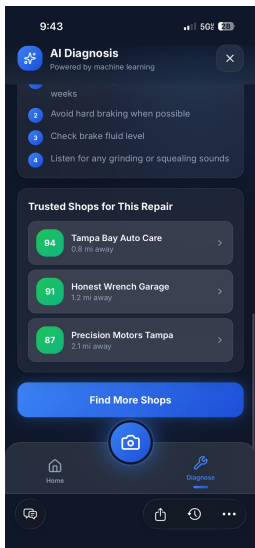
Home & Discovery



AI Diagnose



Results + Pricing



Trusted Shops

“The mechanic space is one instance of a pattern I keep returning to as a PM: information asymmetry creates anxiety, and anxiety creates bad decisions. The design principles here — social trust layers, transparent pricing, multimodal input, AI with explainability — transfer to any domain where users are underserved by the information they need to act confidently.”

— Ruthvik Arepelly, Mechanic Trust